Light

Memory & Storage

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AQA GCSE Computer Science — **Memory and Storage (8525 Paper 2)**. This topic covers the different types of memory within a computer system (RAM, ROM, cache, registers), the need for secondary storage, and the characteristics of magnetic, solid state, and cloud storage solutions.

Exam questions typically ask you to **explain the differences** between types of memory, **describe** why secondary storage is needed, **compare** storage technologies, and **discuss advantages and disadvantages** of cloud storage. The 2024 Paper 2 included questions requiring students to explain volatile and non-volatile memory, and compare storage solutions for real-world scenarios. Expect questions worth 2—6 marks requiring clear, precise explanations.

Why memory and storage matter

Every computer system needs to store data and instructions. Understanding the memory hierarchy is essential for explaining how computers process information efficiently. Different types of memory serve different purposes:

- **Speed vs capacity trade-off:** Faster memory (registers, cache) is expensive and small; slower memory (secondary storage) is cheap and large

- **Volatile vs non-volatile:** Some memory loses data when power is removed; some retains it permanently
- **Main memory vs secondary storage:** The CPU can only directly access main memory; secondary storage holds data long-term
- **Technology choices:** Different storage technologies (magnetic, solid state, cloud) suit different needs and budgets

In the exam, you must be able to explain these concepts precisely using correct terminology. AQA expects you to apply your knowledge to realistic scenarios such as choosing storage for a school, business, or personal device.

What you must know

This checklist covers the key exam-visible concepts from the AQA specification. You should be able to explain each point and apply it to scenarios.

- **Understand** the different types of memory within a computer: RAM, ROM, cache, and registers.
- **Explain** the differences between main memory and secondary storage.
- **Explain** the differences between RAM and ROM, including the terms volatile and non-volatile.
- **Explain** why secondary storage is required.
- **Describe** the operation of solid state and magnetic storage.
- **Discuss** the advantages and disadvantages of solid state and magnetic storage.
- **Explain** the term cloud storage and discuss advantages and disadvantages compared to local storage.

> **Specification note.** AQA 8525 assesses memory and storage as part of systems architecture on Paper 2 (Computing concepts). You must know that main memory is directly accessible by the CPU, while secondary storage is not. AQA uses decimal prefixes: 1 kB = 1000 bytes, 1 MB = 1000 kB, 1 GB = 1000 MB, 1 TB = 1000 GB.

Toolkit

Memory hierarchy at a glance

Memory in a computer is organised in a hierarchy based on speed, capacity, and cost. The closer to the CPU, the faster and more expensive the memory.

Memory type	Speed	Capacity	Cost per byte	Volatility
Registers	Fastest	Smallest (bytes)	Highest	Volatile
Cache	Very fast	Small (KB—MB)	Very high	Volatile
RAM	Fast	Medium (GB)	Medium	Volatile
ROM	Fast	Small (MB)	Medium	Non-volatile

Memory type	Speed	Capacity	Cost per byte	Volatility
Secondary storage	Slow	Large (TB)	Lowest	Non-volatile

Key definitions for the exam

Volatile memory: Memory that loses its contents when the power is switched off. RAM, cache, and registers are volatile.
Non-volatile memory: Memory that retains its contents when the power is switched off. ROM and secondary storage are non-volatile.
Main memory: Memory that is directly accessible by the CPU and used to store data and programs. In AQA exams, this usually refers to RAM and ROM.

Secondary storage: Non-volatile storage that is not directly accessible by the CPU. Used to store data and programs permanently.
Cloud storage: Storage provided at a remote location, accessed via the internet, using magnetic and/or solid state storage at data centres.
Local storage: Storage physically present in or connected to the device, such as internal SSDs or external hard drives.

Storage capacity units (AQA decimal prefixes)

Unit	Symbol	Value in bytes	Relationship
Bit	b	1/8 of a byte	Fundamental unit (0 or 1)
Byte	B	1 byte	8 bits
Kilobyte	kB	1,000 bytes	1000 B
Megabyte	MB	1,000,000 bytes	1000 kB
Gigabyte	GB	1,000,000,000 bytes	1000 MB
Terabyte	TB	1,000,000,000,000 bytes	1000 GB

Exam tip: AQA uses decimal prefixes (1 kB = 1000 bytes), not binary prefixes (1 KiB = 1024 bytes). Always use 1000 as the multiplier in calculations unless the question explicitly states otherwise.

1. Types of memory

A computer contains several types of memory, each serving a specific purpose. Understanding the differences between these memory types is essential for explaining how a computer system operates. The AQA specification requires you to know about four types of memory: RAM, ROM, cache, and registers. Each type has distinct characteristics in terms of speed, capacity, volatility, and purpose.

Memory is organised in a hierarchy. At the top are registers — the fastest but smallest. Next comes cache, then RAM, then ROM. At the bottom is secondary storage, which is the slowest but has the largest capacity. This hierarchy exists because faster memory is more expensive to produce, so computers use small amounts of fast memory for immediate needs and larger amounts of slower memory for less urgent storage.

1.1 RAM (Random Access Memory)

RAM is the main working memory of a computer. It temporarily stores data and instructions that the CPU is currently using or is about to use. When you open an application on your computer, the program code and any data it needs are copied from secondary storage into RAM, where the CPU can access them quickly.

Key characteristics of RAM:

- **Volatile:** Contents are lost when power is switched off
- **Read and write:** Data can be read from and written to RAM
- **Fast access:** Much faster than secondary storage
- **Directly accessible:** The CPU can access any location in RAM directly
- **Temporary:** Stores the operating system, running applications, and data currently in use

When you open an application, it is loaded from secondary storage into RAM so the CPU can execute its instructions quickly. More RAM allows more applications to run simultaneously without slowing down.

The term "random access" refers to the fact that any location in RAM can be accessed directly, without having to read through all the preceding data. This is in contrast to sequential access devices like magnetic tape, where data must be read in order.

RAM capacity is measured in gigabytes (GB). A typical modern computer might have 8 GB, 16 GB, or 32 GB of RAM. The amount of RAM needed depends on what tasks the computer will perform — basic web browsing requires less RAM than video editing or running multiple applications simultaneously.

1.2 ROM (Read Only Memory)

ROM is memory that contains permanent instructions that cannot be changed during normal operation. It stores the basic startup instructions for the computer, known as the BIOS (Basic Input/Output System) or firmware.

Key characteristics of ROM:

- **Non-volatile:** Contents are retained when power is switched off
- **Read only:** Data can be read but not written to during normal use
- **Permanent:** Contents are set during manufacture
- **Stores BIOS/firmware:** Contains the boot instructions that start the computer
- **Small capacity:** Typically only a few megabytes, as it only needs to store essential startup code

When a computer is switched on, the CPU first reads instructions from ROM to perform the Power-On Self-Test (POST) and load the operating system from secondary storage into RAM. Without ROM, the computer would have no instructions to follow when first powered on, and would not know how to start the operating system.

The term "read only" means that under normal circumstances, the data cannot be modified. However, some modern ROM chips (such as EEPROM) can be reprogrammed using special procedures, which allows firmware updates to be installed. This is still called ROM because during normal operation, the memory is only read from.

1.3 Cache memory

Cache is a small amount of very fast memory located between the CPU and RAM. It stores frequently used data and instructions to speed up processing. Cache acts as a buffer, holding copies of data that the CPU is likely to need again soon.

Key characteristics of cache:

- **Volatile:** Contents are lost when power is switched off
- **Very fast:** Faster than RAM, closer to CPU speed
- **Small capacity:** Typically measured in kilobytes or megabytes
- **Expensive:** High cost per byte
- **Automatic:** Managed by hardware, not the user

When the CPU needs data, it first checks the cache. If the data is there (a **cache hit**), it can be accessed very quickly. If not (a **cache miss**), the CPU must fetch it from the slower RAM. Larger cache sizes improve performance because more data can be stored close to the CPU.

Cache works on the principle of locality — programs tend to access the same data and instructions repeatedly, and tend to access data that is stored near recently accessed data. By keeping recently used data in cache, the CPU can avoid the slower process of fetching from RAM.

The effect of cache on performance is significant. Increasing the cache size is one of the factors that can improve CPU performance, alongside clock speed and number of cores.

1.4 Registers

Registers are tiny storage locations inside the CPU itself. They hold the data and instructions that the CPU is currently processing. Registers are the fastest form of memory in a computer system.

Key characteristics of registers:

- **Volatile:** Contents are lost when power is switched off
- **Fastest:** The fastest memory in the computer
- **Smallest:** Capacity measured in bytes (typically 32 or 64 bits each)
- **Inside CPU:** Part of the processor itself
- **Specific purposes:** Different registers hold addresses, data, instructions, or status information

During the Fetch-Execute cycle, registers hold the current instruction, the address of the next instruction, and any data being processed. Without registers, the CPU could not function.

Although the AQA specification states that knowledge of specific registers is not required, you should understand that registers are essential for CPU operation and are the fastest, smallest form of memory.

> #### Worked Example 1 – Identifying memory types
>
> **Question:** State which type of memory (RAM, ROM, cache, or register) matches each description. [4 marks]
>
> (a) Stores the BIOS instructions used to start up the computer.

(b) A small amount of very fast memory that stores frequently accessed data.
(c) The main working memory where the operating system is loaded.
(d) Tiny storage locations inside the CPU that hold data being processed.

Model answer:
(a) ROM **(1)** — ROM stores permanent boot instructions that are retained when power is off.
(b) Cache **(1)** — Cache is small, fast memory between the CPU and RAM.
(c) RAM **(1)** — RAM is the main working memory where programs and data are loaded.
(d) Registers **(1)** — Registers are inside the CPU and hold data currently being processed.

Exam tip: When asked to identify memory types, focus on the key distinguishing feature mentioned. "Boot instructions" = ROM. "Very fast, frequently accessed" = cache. "Main working memory" = RAM. "Inside CPU" = registers.

Worked Example 2 – Volatile vs non-volatile

Question: Explain what is meant by volatile memory. Give one example of volatile memory and one example of non-volatile memory. [3 marks]

Model answer:
Volatile memory is memory that loses its contents when the power is switched off **(1)**.
Example of volatile memory: RAM **(1)** (also accept: cache or registers).
Example of non-volatile memory: ROM **(1)** (also accept: secondary storage, SSD, HDD, flash memory).

Exam tip: The definition of volatile must include the idea of losing contents when power is removed. Do not say "temporary" without explaining what that means.

1.5 Main memory vs secondary storage

It is essential to understand the difference between main memory and secondary storage, as this is frequently tested.

Main memory is memory that is directly accessible by the CPU and used to store data and programs. In AQA exams, this usually refers to:

- RAM — stores running programs and data
- ROM — stores boot instructions

Cache and registers are also directly accessible by the CPU but are typically discussed separately as part of CPU architecture.
Main memory is fast but limited in capacity. RAM is volatile, meaning unsaved work is lost if power fails.

Secondary storage is non-volatile storage that is not directly accessible by the CPU. This includes:

- Hard disk drives (HDD) — magnetic storage
- Solid state drives (SSD) — solid state storage
- Optical discs (CD, DVD, Blu-ray) — optical storage
- USB flash drives — solid state storage

Secondary storage is slower but has much larger capacity. Data must be loaded into RAM before the CPU can process it.

Worked Example 3 – RAM vs ROM comparison

Question: Describe two differences between RAM and ROM. [4 marks]

Model answer:
Difference 1 — Volatility:
RAM is volatile, meaning its contents are lost when power is switched off **(1)**.
ROM is non-volatile, meaning its contents are retained when power is switched off **(1)**.
Difference 2 — Read/write capability:
RAM can be read from and written to during normal operation **(1)**.
ROM can only be read from during normal operation; it cannot be written to **(1)**.

Exam tip: When comparing RAM and ROM, always state both sides clearly. Do not just say "RAM is volatile" — also state "ROM is non-volatile" to show clear comparison.

Worked Example 4 – Why RAM is needed

Question: Explain why a computer needs RAM. [2 marks]

Model answer:
RAM is needed to store the data and instructions that the CPU is currently using or about to use **(1)**.
The CPU can access RAM much faster than secondary storage, which speeds up processing **(1)**.
Alternative second mark: RAM allows multiple programs to be loaded and run at the same time **(1)**.

> **Exam tip:** Focus on the purpose (temporary storage for current use) and the benefit (faster access than secondary storage). Do not just say "to store data" — be specific about what kind of data and why.

2. Secondary storage

Secondary storage provides permanent, non-volatile storage for data and programs. Unlike RAM, secondary storage retains its contents when the computer is switched off. The CPU cannot access secondary storage directly — data must first be loaded into RAM before the CPU can process it. This is a fundamental distinction that you must understand for the exam.

There are several technologies used for secondary storage, but AQA focuses on two main types: magnetic storage (such as hard disk drives) and solid state storage (such as SSDs and USB flash drives). Each technology has different characteristics that make it suitable for different purposes. Understanding these differences allows you to recommend appropriate storage solutions for different scenarios.

2.1 Why is secondary storage needed?

Computers require secondary storage for several important reasons. This is a common exam question, so you should be able to explain the need clearly.

- **Non-volatile storage:** RAM loses its contents when power is removed, so secondary storage is needed to save data permanently
- **Large capacity:** Secondary storage can hold much more data than RAM at a lower cost per byte
- **Store the operating system:** The OS must be stored permanently and loaded into RAM when the computer starts
- **Store applications:** Programs are stored on secondary storage and loaded into RAM when needed
- **Store user files:** Documents, photos, videos, and other files need permanent storage
- **Backup and archive:** Important data can be stored long-term for recovery if needed

2.2 Magnetic storage

Magnetic storage uses magnetised areas on a spinning disk or tape to represent binary data. The most common example is the hard disk drive (HDD), which has been the primary form of secondary storage for decades.

How magnetic storage works:

- Data is stored on circular **platters** coated with magnetic material
- A **read/write head** moves across the spinning platters
- The head magnetises tiny areas to represent 1s and 0s
- To read data, the head detects the magnetic orientation of each area
- Platters spin at high speeds (typically 5400—7200 RPM)

The physical nature of HDDs — with spinning platters and moving heads — means there is a delay while the head moves to the correct position and waits for the right part of the platter to spin underneath. This makes HDDs slower than solid state storage.

Advantages of magnetic storage (HDD):

- Lower cost per gigabyte than solid state storage
- Available in very large capacities (up to several terabytes)

- Well-established, reliable technology

Disadvantages of magnetic storage (HDD):

- Slower read/write speeds than solid state storage
- Moving parts make it vulnerable to physical damage
- Heavier and bulkier than solid state drives
- Higher power consumption
- Produces noise and heat due to spinning platters

2.3 Solid state storage

Solid state storage uses electronic circuits to store data with no moving parts. Examples include solid state drives (SSDs), USB flash drives, and SD cards. The term "solid state" refers to the fact that data is stored using solid electronic components rather than moving mechanical parts.

How solid state storage works:

- Data is stored in **flash memory** chips
- Uses **electrical circuits** to trap electrons in cells
- Each cell can store one or more bits of data
- No moving parts — data is accessed electronically
- Data is retained without power (non-volatile)
- Data can be written and erased electronically

The AQA specification states that students should be aware that SSDs use electrical circuits to persistently store data, but will not need to know precise details such as the use of NAND gates. In simple terms, flash memory works by trapping or releasing electrons in tiny cells, with the presence or absence of electrons representing binary 1s and 0s.

Because there are no moving parts, SSDs can access any data almost instantly — there is no need to wait for a disk to spin or a head to move. This makes them significantly faster than HDDs for both reading and writing data.

Advantages of solid state storage (SSD):

- Much faster read/write speeds than magnetic storage
- No moving parts — more durable and shock-resistant
- Lighter and more compact
- Lower power consumption — better for laptops and mobile devices
- Silent operation — no noise from spinning parts
- Less heat generated

Disadvantages of solid state storage (SSD):

- Higher cost per gigabyte than magnetic storage
- Limited write cycles — cells can wear out over time
- Smaller maximum capacities (though improving)
- Data recovery can be more difficult if the drive fails

2.4 Optical storage

Optical storage uses lasers to read and write data on the surface of a disc. Examples include CDs, DVDs, and Blu-ray discs.

How optical storage works:

- Data is stored as **pits** (indentations) and **lands** (flat areas) on the disc surface
- A laser beam reads the data by detecting how light reflects off the pits and lands
- Writable discs use a laser to burn pits into a special dye layer

Advantages of optical storage:

- Cheap to produce — good for distributing software and media
- Portable and lightweight
- Durable if handled carefully (no moving parts in the disc itself)
- Good for archiving data that rarely needs to be accessed

Disadvantages of optical storage:

- Much slower access speeds than HDDs and SSDs
- Lower capacity than modern HDDs and SSDs
- Discs can be scratched, making data unreadable
- Requires a disc drive, which many modern devices no longer include
- Less commonly used in modern systems

Specification note. AQA 8525 focuses primarily on magnetic and solid state storage, but optical storage (CD, DVD, Blu-ray) may appear in exam questions. You should know the basic operation and be able to compare it with other storage types.

Worked Example 5 — Why secondary storage is needed

Question: Explain why a computer needs secondary storage as well as RAM. [3 marks]

Model answer:
RAM is volatile, so its contents are lost when the computer is switched off **(1)**.
Secondary storage is non-volatile, so it retains data permanently / when power is removed **(1)**.
Secondary storage is needed to store the operating system, applications, and user files so they are available when the computer is next switched on **(1)**.

Exam tip: This question requires you to explain the relationship between RAM and secondary storage. Always mention volatility and give examples of what is stored on secondary storage.

Worked Example 6 — Comparing magnetic and solid state storage

Question: A school is replacing the hard disk drives (HDDs) in its laptops with solid state drives (SSDs). Describe two advantages for the school of using SSDs instead of HDDs. [4 marks]

Model answer:
Advantage 1: SSDs are faster than HDDs **(1)**, so laptops will boot up more quickly and applications will load faster, improving productivity for students and staff **(1)**.

Advantage 2: SSDs have no moving parts, making them more durable **(1)**, so the laptops are less likely to be damaged if dropped by students, reducing repair costs **(1)**.

Exam tip: When asked for advantages in a scenario, always link the advantage to the context. Do not just say "SSDs are faster" — explain why faster is beneficial for a school.

Worked Example 7 – How magnetic storage works

Question: Describe how data is stored on a hard disk drive. [3 marks]

Model answer:
Data is stored on circular platters coated with magnetic material **(1)**.
A read/write head moves across the spinning platters **(1)**.
The head magnetises tiny areas on the platter surface to represent binary 1s and 0s **(1)**.

Exam tip: AQA expects you to know the basic operation of magnetic storage. Key terms: platters, read/write head, magnetised areas, spinning. You do not need to explain the physics in detail.

Worked Example 8 – How solid state storage works

Question: Describe how data is stored on a solid state drive. [2 marks]

Model answer:
Data is stored in flash memory chips / using electrical circuits **(1)**.
Electrons are trapped in cells to represent binary data / there are no moving parts **(1)**.

Exam tip: AQA states that students should be aware that SSDs use electrical circuits to persistently store data but will not need to know precise details such as the use of NAND gates. Keep your answer simple and focused on "electrical circuits" and "no moving parts".

3. Cloud storage and comparisons

Cloud storage has become increasingly important in modern computing. It represents a shift from storing data on physical devices that you own to storing data on remote servers managed by third parties. AQA expects you to understand what cloud storage is,

how it differs from local storage, and be able to discuss its advantages and disadvantages in realistic scenarios.

In exams, you may be asked to recommend whether an individual or organisation should use cloud storage, local storage, or a combination of both. To answer these questions well, you need to understand the characteristics of each approach and be able to apply them to specific contexts.

3.1 What is cloud storage?

Cloud storage is storage provided at a remote location, typically in large data centres, and accessed via the internet. The physical storage at these data centres uses magnetic and/or solid state storage technology — the same technologies as local storage, just located elsewhere and managed by a service provider.

The term "cloud" refers to the internet and the abstract nature of the storage from the user's perspective. Users do not need to know where their data is physically stored or how it is managed — they simply access it through the internet as if it were floating in a "cloud".

Key features of cloud storage:

- Data is stored on servers at remote locations
- Accessed via an internet connection
- Managed by a third-party provider (e.g., Google, Microsoft, Amazon)
- Users pay for storage capacity or subscribe to a service
- Data is typically replicated across multiple servers for reliability

Examples of cloud storage services:

- Google Drive
- Microsoft OneDrive
- Apple iCloud
- Dropbox
- Amazon Web Services (AWS)

3.2 Local storage

Local storage refers to storage devices that are physically present in or directly connected to the user's device.

Examples of local storage:

- Internal hard disk drives (HDD)
- Internal solid state drives (SSD)
- External hard drives
- USB flash drives
- Memory cards (SD cards)

Key features of local storage:

- Physically accessible to the user
- No internet connection required to access data
- User has direct control over the storage device
- One-time purchase cost (no ongoing subscription)
- Data security is the user's responsibility

3.3 Advantages and disadvantages of cloud storage

Advantages of cloud storage:

- **Access from anywhere:** Files can be accessed from any device with an internet connection
- **Automatic backup:** Data is backed up automatically by the provider, reducing risk of data loss
- **Easy sharing:** Files can be shared with others easily via links or permissions
- **Scalability:** Storage capacity can be increased easily by paying for more
- **Collaboration:** Multiple users can work on the same files simultaneously
- **No hardware maintenance:** The provider maintains the servers and storage devices
- **Disaster recovery:** Data is protected if the user's local device is lost, stolen, or damaged

Disadvantages of cloud storage:

- **Requires internet:** Files cannot be accessed without an internet connection
- **Security concerns:** Data is stored on third-party servers, potentially vulnerable to hacking
- **Privacy concerns:** The provider may have access to user data
- **Ongoing costs:** Subscription fees can add up over time
- **Dependency on provider:** If the provider has an outage or goes out of business, data may be inaccessible
- **Slower access:** Upload and download speeds depend on internet bandwidth
- **Data transfer limits:** Some providers limit how much data can be transferred

Worked Example 9 — Defining cloud storage

Question: Explain what is meant by cloud storage. [2 marks]

Model answer:
Cloud storage is storage provided at a remote location / on servers in data centres **(1)**.
It is accessed via the internet / uses magnetic and/or solid state storage at the remote location **(1)**.

Exam tip: The AQA specification states that cloud storage "uses magnetic and/or solid state storage at a remote location." Include the idea of remote storage and internet access in your definition.

Worked Example 10 — Cloud vs local storage scenario

Question: A photographer stores all their photos on a local external hard drive. Discuss the advantages and disadvantages of using cloud storage instead. [6 marks]

Model answer (banded response):
Advantages of cloud storage:
The photographer could access their photos from anywhere using any device with an internet connection, which would be useful when meeting clients or travelling **(1)**.

Cloud storage provides automatic backup, so if the photographer's computer or external drive was lost, stolen, or damaged, their photos would still be safe **(1)**.
Photos could be shared easily with clients by sending a link, rather than having to physically transfer files **(1)**.
Disadvantages of cloud storage:
The photographer would need a reliable internet connection to upload and download large photo files, which may be slow or unavailable in some locations **(1)**.
There may be ongoing subscription costs, whereas the external hard drive is a one-time purchase **(1)**.
Storing photos with a third party raises security and privacy concerns, especially if the photos are commercially sensitive or include clients **(1)**.

Exam tip: For "discuss" questions, provide balanced points covering both advantages and disadvantages. Link each point to the scenario (photographer, photos, clients) for full marks.

Worked Example 11 — Extended response on storage choice

Question: A small business is deciding whether to use solid state drives (SSDs) or hard disk drives (HDDs) for storing customer records. Discuss the factors the business should consider. [6 marks]

Model answer (Level 3 response, 5—6 marks):
The business should consider several factors when choosing between SSDs and HDDs for customer records.
Speed: SSDs are significantly faster than HDDs, which would allow staff to access customer records more quickly, improving efficiency and customer service. However, for storing records that are not accessed frequently, speed may be less important.
Cost: HDDs are cheaper per gigabyte than SSDs. If the business has a large volume of records and a limited budget, HDDs may be more cost-effective. However, the price difference has reduced in recent years.
Reliability: SSDs have no moving parts, making them more durable and less likely to fail due to physical damage. For important customer records, the increased reliability of SSDs may justify the higher cost. HDDs can be damaged if dropped or knocked.
Capacity: HDDs are available in larger capacities at lower prices. If the business needs to store a very large amount of data, HDDs may be necessary.
Conclusion: For frequently accessed customer records where speed and reliability are priorities, SSDs would be the better choice despite the higher cost. For archiving older records that are rarely accessed, HDDs may be sufficient and more economical.

Exam tip: Extended response questions are marked in bands. To reach the highest band, provide a structured discussion that considers multiple factors, applies them to the scenario, and reaches a reasoned conclusion.

Worked Example 12 — Storage capacity calculation

Question: A video file has a size of 2.5 GB. How many of these video files could be stored on a 1 TB hard drive? Show your working. [3 marks]

Model answer:
Step 1: Convert TB to GB.
1 TB = 1000 GB **(1)**
Step 2: Divide the storage capacity by the file size.
1000 GB ÷ 2.5 GB = 400 **(1)**
Answer: 400 video files could be stored on the hard drive **(1)**.

Exam tip: Always use AQA's decimal prefixes: 1 TB = 1000 GB (not 1024 GB). Show your working clearly to gain method marks even if your final answer is incorrect.

Worked examples summary

This topic includes 12 worked examples covering all key areas tested by AQA. Use these as templates for structuring your own exam answers.

Example	Topic	Question type	Marks
1	Identifying memory types	State / identify	4
2	Volatile vs non-volatile	Explain + examples	3
3	RAM vs ROM comparison	Describe differences	4
4	Why RAM is needed	Explain	2
5	Why secondary storage is needed	Explain	3
6	SSD vs HDD advantages	Describe in context	4
7	How magnetic storage works	Describe	3
8	How solid state storage works	Describe	2
9	Defining cloud storage	Explain	2
10	Cloud vs local storage	Discuss (scenario)	6

Example	Topic	Question type	Marks
11	SSD vs HDD for business	Discuss (extended)	6
12	Storage capacity calculation	Calculate	3

Quick reference

Memory types comparison

Feature	RAM	ROM	Cache	Registers
Volatility	Volatile	Non-volatile	Volatile	Volatile
Read/Write	Read and write	Read only	Read and write	Read and write
Speed	Fast	Fast	Very fast	Fastest
Typical capacity	GB	MB	KB—MB	Bytes
Location	Main memory	Main memory	Between CPU and RAM	Inside CPU
Purpose	Store running programs and data	Store boot instructions (BIOS)	Store frequently used data	Hold data being processed

Secondary storage comparison

Feature	Hard Disk Drive (HDD)	Solid State Drive (SSD)	Optical Disc (CD/DVD/Blu-ray)
Technology	Magnetic (spinning platters)	Solid state (flash memory)	Optical (laser reads pits/lands)
Moving parts	Yes (platters, read/write head)	No	Yes (spinning disc, laser)
Speed	Medium	Fastest	Slowest
Cost per GB	Low	Higher	Very low (for distribution)
Durability	Vulnerable to physical damage	Most durable (shock-resistant)	Discs can be scratched
Typical capacity	500 GB — several TB	128 GB — 4 TB	700 MB (CD) — 50 GB (Blu-ray)

Feature	Hard Disk Drive (HDD)	Solid State Drive (SSD)	Optical Disc (CD/DVD/Blu-ray)
Portability	Medium (external drives)	High (small, lightweight)	High (discs are thin and light)
Common use	Mass storage, backups	Operating system, applications	Software distribution, media

Cloud storage vs local storage

Feature	Cloud storage	Local storage
Location	Remote data centres	On the device or connected to it
Access	Requires internet connection	No internet required
Cost	Ongoing subscription	One-time purchase
Backup	Automatic by provider	User's responsibility
Sharing	Easy via links/permissions	Requires physical transfer
Security control	Managed by provider	Controlled by user
Access from multiple devices	Yes	Limited
Scalability	Easy to increase capacity	Requires buying new hardware

Key definitions for the exam

Term	Definition (mark-scheme style)
Volatile memory	Memory that loses its contents when the power is switched off.
Non-volatile memory	Memory that retains its contents when the power is switched off.
Main memory	Memory directly accessible by the CPU; in AQA exams this means RAM and ROM.
Secondary storage	Non-volatile storage that is not directly accessible by the CPU.
RAM	Volatile memory that stores data and programs currently in use.

Term	Definition (mark-scheme style)
ROM	Non-volatile memory that stores permanent instructions (e.g., BIOS).
Cache	Small, fast memory between the CPU and RAM storing frequently used data.
Registers	Tiny, fast storage locations inside the CPU holding data being processed.
Cloud storage	Storage at a remote location accessed via the internet using magnetic and/or solid state storage.
Magnetic storage	Storage using magnetised areas on spinning platters (e.g., HDD).
Solid state storage	Storage using electrical circuits with no moving parts (e.g., SSD).
Optical storage	Storage using lasers to read pits and lands on a disc surface (e.g., CD, DVD, Blu-ray).

Typical pitfalls

Pitfall 1 — Confusing RAM with secondary storage.
Wrong: "RAM stores files permanently."
Right: RAM is volatile and loses data when power is off. Secondary storage (HDD/SSD) stores files permanently.

Pitfall 2 — Saying ROM is "secondary storage".
Wrong: "ROM is a type of secondary storage."
Right: ROM is main memory (directly accessible by CPU). It stores boot instructions, not user files.

Pitfall 3 — Using binary prefixes instead of decimal.
Wrong: "1 kB = 1024 bytes."
Right: AQA uses decimal prefixes: 1 kB = 1000 bytes, 1 MB = 1000 kB, etc.

Pitfall 4 — Confusing cache with secondary storage.
Wrong: "Cache is where you store your files."
Right: Cache is small, fast memory that temporarily stores frequently accessed data. It is volatile and managed automatically by hardware.

Pitfall 5 — Saying "memory" when you mean "storage".
Wrong: Using "memory" and "storage" interchangeably.

Right: Memory (RAM, ROM, cache, registers) is directly accessible by the CPU. Storage (HDD, SSD) is not directly accessible and is used for permanent data.

Pitfall 6 — Incomplete comparison answers.
Wrong: "RAM is volatile" without stating "ROM is non-volatile."
Right: Always state both sides when comparing: "RAM is volatile (loses data when power is off), while ROM is non-volatile (retains data when power is off)."

Pitfall 7 — Forgetting to link advantages to the scenario.
Wrong: "SSDs are faster."
Right: "SSDs are faster, so the school laptops will boot up more quickly, improving lesson time."

Pitfall 8 — Saying cloud storage is "not physical".
Wrong: "Cloud storage is virtual, not physical."
Right: Cloud storage uses physical magnetic and/or solid state storage at remote data centres. The data is real; it is just stored somewhere else.

Pitfall 9 — Confusing volatile with "easily lost".
Wrong: "Volatile means the data can be deleted easily."
Right: Volatile means data is lost specifically when power is removed. It has nothing to do with ease of deletion.

Pitfall 10 — Forgetting registers exist.
Wrong: Listing only RAM, ROM, and cache when asked about memory types.
Right: The four types of memory in a computer are: RAM, ROM, cache, and registers. Registers are inside the CPU and are the fastest type of memory.

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